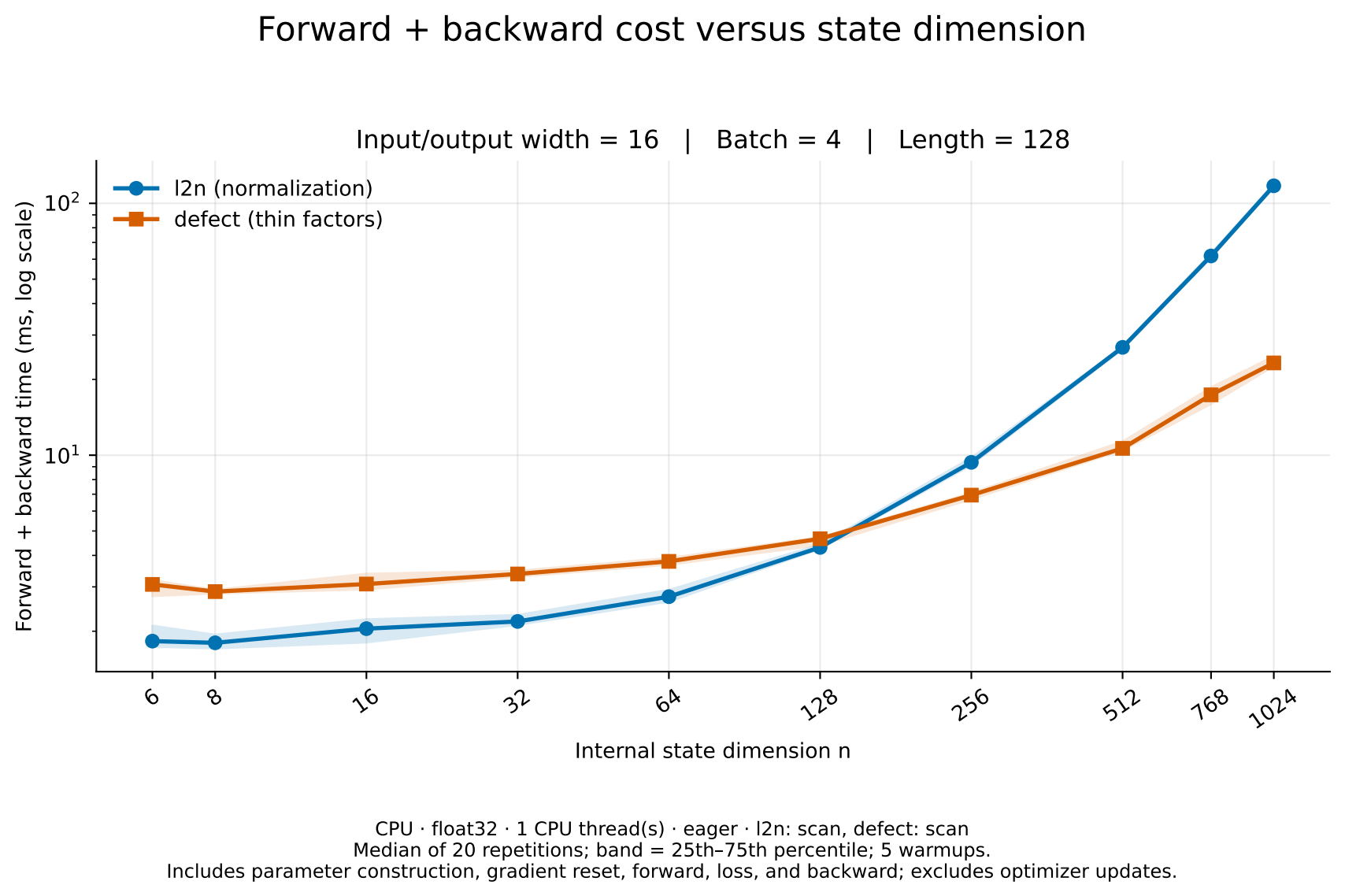
Forward + backward cost versus state dimension
Input/output width = 16 | Batch = 4 | Length = 128
l2n (normalization)
defect (thin factors)
Forward + backward time (ms, log scale)
10^2
10^1
6
8
16
32
64
128
256
512
768
1024
Internal state dimension n
CPU · float32 · 1 CPU thread(s) · eager · l2n: scan, defect: scan
Median of 20 repetitions; band = 25th–75th percentile; 5 warmups.
Includes parameter construction, gradient reset, forward, loss, and backward; excludes optimizer updates.